Clinical & Experimental Ophthalmology

WILEY

Clinical & Experimental Ophthalmology

ORIGINAL ARTICLE-CLINICAL SCIENCE OPEN ACCESS

Development and Validation of a Deep Learning System for the Provision of a District-Wide Diabetes Retinal Screening Service

Jason R. Daley[1,2] | Xingdi Wang[1,2] | Natalie Ngo[1,3] | Chee L. Khoo[3,4] | Peter Heydon[1,2,3,5] | Gerald Liew[2,6,7] | Vallimayil Vallayutham[2,8,9] | Tobias Kongbrailatpam[10] | Uchechukwu L. Osuagwu[10] | Marko Andric[8] | Wei Xuan[3,11] | David Simmons[8,10] | Shweta Kaushik[2,6,8,10]

[1]Liverpool Hospital, Sydney, Australia | [2]The University of Sydney, Sydney, Australia | [3]The University of New South Wales, Sydney, Australia | [4]Healthfocus Family Practice, Sydney, Australia | [5]Royal Prince Alfred Hospital, Sydney, Australia | [6]Westmead Hospital, Sydney, Australia | [7]Westmead Institute for Medical Research, Sydney, Australia | [8]Campbelltown Hospital, Sydney, Australia | [9]The Children's Hospital at Westmead, Sydney, Australia | [10]Western Sydney University, Sydney, Australia | [11]Ingham Institute for Applied Medical Research, Sydney, Australia

Correspondence: Shweta Kaushik (shweta.kaushik@sydney.edu.au)

Received: 8 February 2025 | **Revised:** 3 May 2025 | **Accepted:** 8 May 2025

Funding: This work was supported by Ophthalmic Research Institute of Australia.

Keywords: artificial intelligence | deep learning | diabetic retinopathy | dual-modality | screening

ABSTRACT

Background: Artificial intelligence (AI) enhanced retinal screening could reduce the impact of diabetic retinopathy (DR), the leading cause of preventable blindness in Australia. This study assessed the performance and validity of a dual-modality, deep learning system for detection of vision-threatening diabetic retinopathy (vtDR) in a multi-ethnic community.

Methods: Cross-sectional (algorithm-validation) study with the deep learning system assessing fundus photographs for gradability and severity of DR, and optical coherence tomography (OCT) scans for diabetic macular oedema (DMO). Internal validation of each algorithm was performed using a computer-randomised 80:20 split. External validation was by comparison to standard grading provided by two ophthalmologists in 748 prospectively recruited persons with diabetes (age $\geq$ 10) from hospital diabetes clinics and a general practice. Main outcome measures included sensitivity, specificity and the area under the receiver operating characteristic curve (AUC).

Results: Internal validation revealed robust test characteristics. When compared to ophthalmologists, the system achieved an AUC of 0.92 (95% CI 0.90–0.94) for fundus photograph gradeability, 0.91 (95% CI 0.85–0.94) for the diagnosis of severe non-proliferative DR/proliferative DR and 0.90 (95% CI 0.87–0.96) for DMO detection from OCT scans. It demonstrated real-world applicability with an AUC of 0.94 (95% CI 0.91–0.97), sensitivity of 92.7% and specificity of 95.5% for detection of vtDR. Ungradable images occurred in 55 participants (7.4%).

Conclusions: The dual-modality, deep learning system can diagnose vtDR from fundus photographs and OCT scans with high levels of accuracy and specificity. This could support a novel model of care for DR screening in our community.

1 | Introduction

Diabetic retinopathy (DR) is the most common microvascular complication of diabetes and the leading cause of preventable blindness in the working population worldwide [1]. As of 2024, the prevalence of diabetes in Australia was estimated at 1.5 million (5.9%) with a projected 2.5-fold increase in the next two decades [2]. Diabetes and DR impose a significant burden on the healthcare system, disproportionately affecting certain geographical areas and population demographics [3]. One such area is South Western Sydney (SWS), which reports a diabetes prevalence of 6.9%, one of the highest rates in Australia [4]. SWS is characterised by a higher proportion of Indigenous Australians and migrant populations, along with lower average income and education levels compared to the national average [5]. Eye health services are limited in this district. The 2023 Liverpool Eye and Diabetes Study (LEADS) by Liew et al. found that 52.3% of adults with diabetes, attending the largest of two public ophthalmology clinics in SWS, had any degree of DR, while 31.5% had vision-threatening diabetic retinopathy (vtDR) [4, 5]. Adults requiring public ophthalmic care, wait up to 12 months for a non-urgent appointment. This highlights the unmet clinical need for a diabetes retinal screening service, which could enable early diagnosis, timely treatment and prevention of vision loss [6].

Currently, the Royal Australian and New Zealand College of Ophthalmologists (RANZCO) Screening and Referral Pathway for DR, requires a visual acuity assessment and fundus photography at least every 2 years, which is reviewed by an optometrist or general practitioner (GP) and referred to an ophthalmologist if required [7]. In Australia, DR screening takes place in a wide range of public and private health settings, including optometry/ophthalmology clinics, general practices, Aboriginal Community Controlled Health Organisations (ACCHOs), visiting outreach services and telehealth [8]. However, this pathway imposes substantial financial and workforce challenges, especially in rural and remote areas, where there are fewer eye practitioners to provide DR screening. In addition, the majority of persons with diabetes are managed in primary care by GPs, but DR screening is rarely conducted in this setting [9]. Studies indicate that only 50%–77% of non-Indigenous Australians and 20%–44% of Indigenous Australians undergo appropriate DR screening. This disparity is driven by multiple factors, including limited patient education, travel barriers, operational costs, time constraints, insufficient Medicare reimbursement and lack of confidence in GP interpretation of retinal images [10–14]. Of late, there has been a trend towards the use of artificial intelligence (AI) and deep learning algorithms (DLAs) to provide DR screening as a cost, time and labour-effective solution. This has largely been driven by increases in computational power and the availability of big data methodology. DLAs with convolutional neural networks (CNNs) can learn and detect complex features based on retinal images and diagnose DR with specialist-level performance, despite using different methods from human clinicians. CNNs utilise multiple filters and layers to identify colours, edges and lines in a hierarchical structure to then comprehend the entire image. The United States Food and Drug administration (US FDA) has provided a recommendation for a minimum sensitivity of 80% and specificity of 82.5% for DLAs that are capable of diagnosing DR [15]. There has been only one deep learning system that has incorporated both fundus photography and optical coherence tomography (OCT) in detecting and grading DR [16].

The primary aim of this study was to assess the diagnostic performance and clinical validity of a dual-modality, deep learning system in detecting vtDR. The secondary aim was to report the overall prevalence of DR among the South Western Eye and Diabetes Deep Learning Algorithm (SWEDDLA) study participants. Given that the majority of people with diabetes are managed in primary care, our deep learning system has the potential to address a significant gap in current service delivery. A clinically validated system within a multi-ethnic community could drive meaningful improvements in DR screening efficiency, accuracy and patient outcomes by enabling automated, point-of-care image interpretation. Integrating this technology into general practices, optometry clinics and diabetes clinics aligns with recent trends for AI-driven solutions and supports national efforts to detect persons requiring specialist-level care early, thereby preventing vision loss.

2 | Methods

The SWEDDLA study was a cross-sectional, algorithm-validation study conducted from June 2021. The SWEDDLA study was approved by the South Western Sydney Local Health District Human Research Ethics Committee (protocol number: 2020/ETH01693) and adhered to the principles outlined in the Declaration of Helsinki. Informed, written consent was obtained from each study participant or guardian.

2.1 | Recruitment and Screening

Consecutive persons with diabetes, attending the adult and paediatric diabetes clinics at Campbelltown Hospital, the Western Sydney University Campbelltown hospital site-located interprofessional learning (IPL) diabetes clinic, the Liverpool Hospital high-risk foot and diabetes clinic and the Healthfocus family practice, were screened with a macular and optic disc-centred fundus photograph plus an OCT scan for the presence and severity of DR. All types of diabetes were included, as long as the diagnosis was based on established criteria such as blood glucose concentration, glycated haemoglobin and/or the oral glucose tolerance test. All participants were aged 10 years or older. We excluded patients who were unable to give informed consent or were medically unstable. Recruitment occurred across all sites between June 2021 and July 2023. Table-top retinal cameras (Zeiss Cirrus Photo 600w/R&A, Jena, Germany) were used at each study site by a trained orthoptist. However, at the Healthfocus family practice, a portable hand-held fundus camera (Welch-Allyn Retinavue 100 Imager Pro, New York, USA), without OCT capability, was operated by the GP. Participants were dark-adapted for 5 min if they were under the age of 55 and dilated with tropicamide 0.5% if they were over 55, to maximise the gradability of images.

2.2 | Deep Learning System Development

Each DLA was created using an open-source, code-free, cloud-based, AI programme, employing specialised CNNs for image feature recognition [17, 18]. The algorithms for assessing fundus photograph gradability and DR severity were developed using EfficientNet (Google AI., California, USA), while the OCT algorithm was built on TensorFlow's Inception-v3 (Alphabet Inc., California, USA) CNN architecture, both of which have demonstrated superior performance [17]. The DLAs were created from a dataset of 55 261 fundus photographs and 16 000 OCT scans, obtained from publicly available retinal image datasets. The fundus photographs were captured using different cameras from a multi-ethnic cohort, with varying severities of DR based on the International Clinical Diabetic Retinopathy Severity scale. Despite being in different digital formats, all images were compatible with and successfully analysed by our programme. These images were then regraded according to a modified Scottish Diabetic Retinopathy Scheme [19]. We introduced a modification where fundus photographs showing any signs of laser photocoagulation were graded as severe non-proliferative diabetic retinopathy (NPDR), unless there was neovascularisation, where we graded it as proliferative diabetic retinopathy (PDR). Additionally, maculopathy was not assessed using fundus photography but instead evaluated through OCT. A gradable fundus photograph was defined as being able to visualise two-thirds of the retinal field and the third-generation vascular branches within one optic disc diameter to the fovea and optic nerve head. vtDR was defined as severe NPDR/PDR on the fundus photograph and/or diabetic macular oedema (DMO) on the OCT scan. Non-vtDR was defined as normal images, those with mild or moderate NPDR and no DMO. The OCT images were graded for the presence or absence of DMO. DMO was defined as the presence of intraretinal fluid, represented as one or more hyporeflective, cystoid spaces within the field of the macular five-line raster in one or more slices, that was large enough to change the thickness of the retina, distort individual layers of the retina or compress overlying inner retinal layers.

We created a combined deep learning system, consisting of three DLAs, each used to predict a different aspect of the image: the gradability of the fundus photograph, the presence of severe NPDR/PDR in the fundus photograph and the detection of DMO on the OCT scan. Each DLA outputted a value between 0 and 1, which indicated its prediction. A value closer to 0 was indicative of a negative result, for example, ungradable fundus photograph, no severe NPDR/PDR and no DMO. Similarly, if the value was at or above the operating point of 0.50, it was indicative of a positive result, for example, gradable fundus photograph, severe NPDR/PDR and DMO. Data augmentation techniques, implemented using Python (Python Software Foundation, Delaware, USA), expanded the dataset and enhanced performance. These algorithms were integrated to develop a deep learning system called Diabetic Retinopathy OCT Open souRce Artificial Intelligence (DOCTOR AI), hereafter referred to as the deep learning system. Thus, an image is uploaded, evaluated separately by each algorithm with a final analysis generated, and any abnormal results are flagged to the tester. A heat map is also generated, enabling the tester to visualise the areas contributing to the DLAs diagnosis. An example of the user interface is shown below in Figure 1.

The algorithm was trained using 80% of the images (computer randomised) and validated against the remaining 20% within a hold-out test set. External validation was performed using routinely collected, de-identified fundus photographs and OCT scans from the recruited population, and comparison was made with the standard grading pathway provided by two consultant ophthalmologists experienced in DR management. In cases of disagreement or diagnostic uncertainty, the final grade was determined through consensus discussion. The primary clinical endpoint was vtDR, defined as either severe NPDR/PDR on fundus photography and/or DMO on the OCT scan.

2.3 | Statistical Analysis

We performed sample size calculations based on nomograms created by Malhotra and Indrayan for use in diagnostic medical tests [20]. It demonstrated that a minimum of 1400 eyes (700 participants) were required for a diagnostic study to produce a sensitivity and specificity of 90%, assuming absolute precision (*p* value) of 0.05 and disease prevalence of 10%. The accuracy, sensitivity, specificity, area under the receiver operating characteristic curve (AUC), positive and negative predictive values (PPV and NPV) of each algorithm was calculated, allowing assessment of the algorithm's performance in comparison to the grading of DR by ophthalmologists. Fundus and OCT gradability was defined as the percentage of having at least one gradable photograph/scan per person. All data were statistically analysed using SAS software (SAS, v9.4, North Carolina, USA) except the AUC graphs and calculations, which were generated using R (version 4.4.2, Auckland, New Zealand).

3 | Results

3.1 | Demographics

Overall, 748 participants including 87 children (age < 18 years) were recruited across study sites: 479 (64%) from Campbelltown Hospital Diabetes Clinics, 207 (27.7%) from Healthfocus Family Practice at Ingleburn, 42 (5.6%) from Liverpool Hospital high-risk foot and diabetes clinics, and 20 (2.7%) from the Western Sydney University Campbelltown Hospital site-located IPL Clinic. There were 2730 fundus photographs and 1583 OCT scans. Participant demographics, systemic risk factors and diabetes history were summarised in Table 1. The median duration of diabetes was 9.5 years (range 0–62 years).

vtDR was found in 63 participants (8.4%), DMO in 42 participants (7.8%) and ungradable images in only 55 participants (7.4%). There were five cases of moderate NPDR in the paediatric participants, but no vtDR. The overall prevalence of each DR grade is displayed in Table 2.

3.2 | Internal Validation

The internal validation test characteristics of each DLA were calculated, including accuracy, sensitivity, specificity, PPV, NPV and AUC, and were summarised in Table 3. This was performed with a computer-randomised 80:20 split of the training images. Three DLAs were tested. The first algorithm determined the gradeability of the fundus photograph, the second algorithm diagnosed severe NPDR/PDR, and the third algorithm detected the presence or absence of DMO on the OCT scan. The AUC values for these algorithms were plotted on Figure 2.

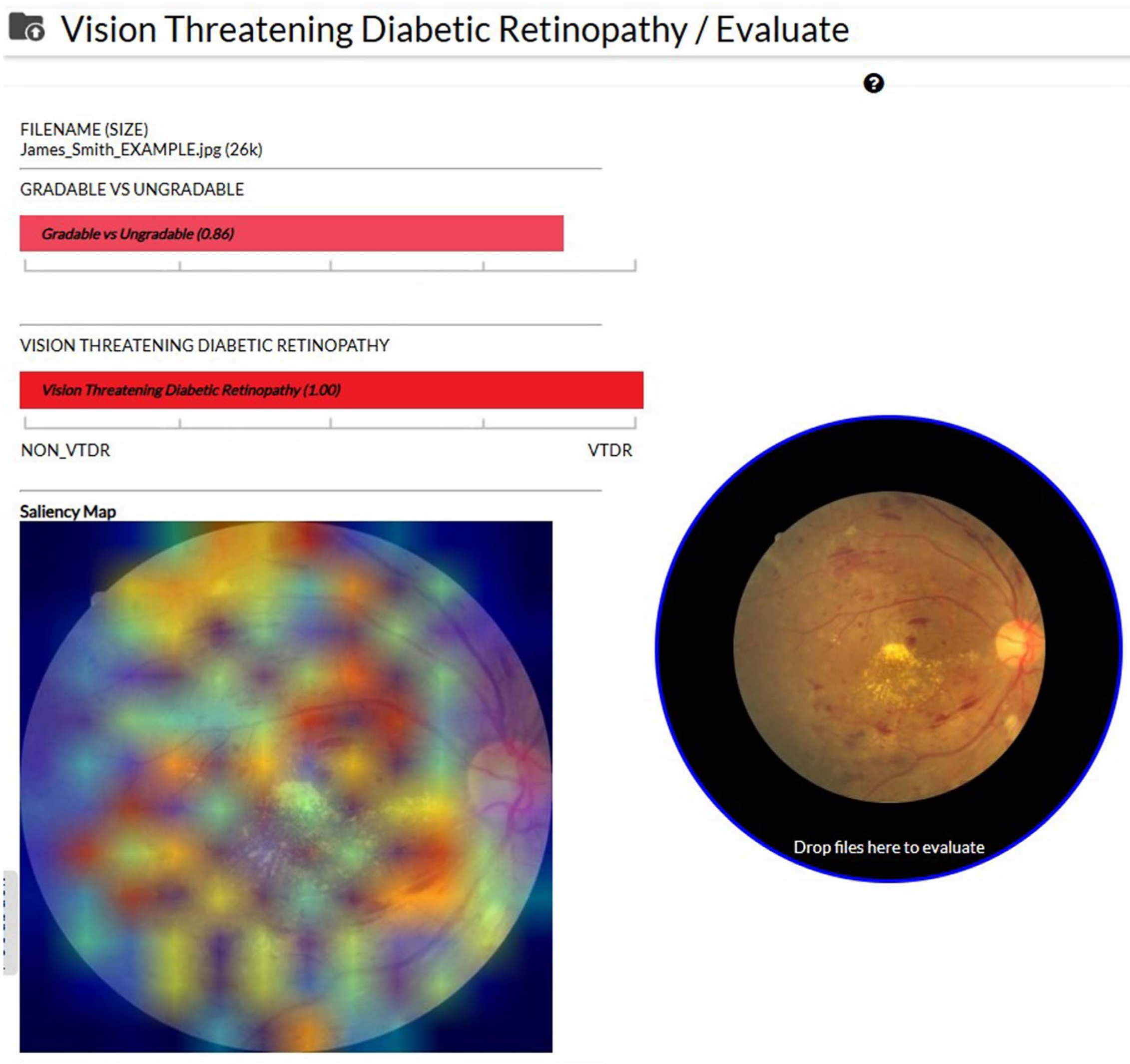

FIGURE 1 | User interface of the deep learning system. It shows an example fundus photograph of a patient with diabetic retinopathy. The deep learning system has identified this as a gradable photograph with 0.86 probability and diagnosed vision-threatening diabetic retinopathy with a 1.0 probability. A saliency (heat) map demonstrates the areas used to make the system's diagnosis.

3.3 | External Validation

Each of the three DLAs was externally validated on the prospectively recruited participants, as shown in Table 4, with their AUC values plotted in Figure 3. For the detection of the primary clinical endpoint of vtDR, we combined the performance of the severe NPDR/PDR and DMO OCT algorithms. The definition of vtDR was severe NPDR/PDR on fundus photography and/or DMO on the OCT scan. The deep learning system was able to achieve an AUC of 0.94 (95% CI 0.91–0.97), accuracy of 95.2%, sensitivity of 92.7%, specificity of 95.5%, PPV of 67.9% and NPV of 99.2% for the diagnosis of vtDR from fundus photography and OCT scans. When both modalities were considered, the deep learning system missed no cases of severe NPDR/PDR and only one case of DMO on the OCT in a participant who did not have fundus photography.

Fundus gradability was overall very high. For the table-top retinal cameras (Zeiss Cirrus Photo 600w/R&A, Jena, Germany), fundus gradability was 92.6% and OCT gradability was 98.8%. The fundus gradability was lower for the portable retinal camera (Welch-Allyn Retinavue 100 Imager Pro, New York, USA), at 80.7%.

4 | Discussion

In the SWEDDLA study, we developed a dual-modality, deep learning system that diagnosed fundus photograph gradability, severe NPDR/PDR and DMO, with similar performance to consultant ophthalmologists within a real-life diabetes retinal screening service. For detecting vtDR from fundus photographs and OCT scans, our system has surpassed the US FDA recommended test characteristics, with an AUC of 0.94, accuracy of 95.2%, sensitivity of 92.7% and specificity of 95.5% [21]. This study was conducted using a large external validation dataset that included 748 adult and paediatric participants from our local SWS area. When compared to other deep learning systems that have been prospectively evaluated, our system demonstrated high accuracy and specificity, with acceptable sensitivity.

TABLE 1 | Demographics of the South Western Eye and Diabetes Deep Learning Algorithm (SWEDDLA) study participants.

Demographics	
Number of participants	748
Number of eyes examined	1482
Ethnicity (*n*, %)	Oceanian (320 [42.8%])
	European (99 [13.3%])
	Asian (89 [11.9%])
	African and Middle Eastern (34 [4.5%])
	Aboriginal or Torres Strait Islander (18 [2.4%])
	People of Americas (10 [1.3%])
	Prefer not to say (19 [2.5%])
	Not reported (159 [21.3%])
Age (mean ± SD, range years)	44.3 ± 21.0, (10–92)
Sex, *n* (%)	
Male	364 (48.7%)
Female	384 (51.3%)
Systemic risk factors	
Body-mass index, kg/m^2 (mean ± SD)	29.9 ± 7.9
Heart attack/stroke history, *n* (%)	108 (14.4%)
Smokers, *n* (%)	272 (37.1%)
Blood pressure, mmHg (mean ± SD, mm Hg)	
Systolic	127 ± 18.5
Diastolic	76 ± 11.1
Diabetes type, *n* (%)	
Type 1	274 (36.6%)
Type 2	435 (58.2%)
Gestational[a]	29 (3.9%)
Other	10 (1.3%)
HbA1c (mean % ± SD)	8.1 ± 2.2
Median diabetes duration, years, (range)	9.5 (0–62)

Abbreviation: HbA1c, glycated haemoglobin.
[a]Includes overt diabetes in pregnancy.

Past studies have demonstrated wide-ranging and variable performance of single-modality DLAs. In 2016, Gulshan et al. [22], prospectively evaluated a pioneering DLA, supported by Google, in multiple primary care sites within Thailand. For the detection of vtDR, their algorithm had a sensitivity of 91.4% and a specificity of 95.4% [23]. Similarly, EyeART, developed by Ipp et al. [24] in the United Kingdom, was the first DLA to gain US FDA approval for the classification of vtDR. It was prospectively validated in a study of 893 persons with diabetes, achieving a sensitivity of 95.1% and specificity of 89.0% for the detection of vtDR [24]. A multicentre, head-to-head, real-world validation study of seven automated, AI DR screening algorithms found substantial differences in overall performance [25]. Using an arbitrated data set as the ground truth, sensitivities ranged from 51.0% to 85.9% and specificities from 60.4% to 83.7% [25]. Similarly, a 2024 meta-analysis of 34 prospective, real-world validation studies found an overall pooled accuracy of 81%, sensitivity of 94% and specificity of 89% for the diagnosis of DR by DLAs [26]. The wide-ranging, variable performance of these DLAs highlights the need for rigorous validation on large, prospectively recruited and real-world populations before clinical implementation.

DR screening algorithms must prioritise diagnostic performance and safety, while minimising unnecessary workload, a challenge that could be addressed by adopting a dual-modality approach with fundus photography and OCT scans. For example, Shah et al. evaluated the performance of a DLA for detecting vtDR in a screening programme involving 2680 persons with diabetes. The algorithm achieved a sensitivity of 100% and specificity of 95% but its PPV was only 33% [27]. These results would not be ideal in primary care settings, due to the relatively low expected PPV, which could lead to unnecessary referrals for healthy individuals, or those with non vision-threatening disease [28]. The simultaneous use of OCT could address this limitation by enhancing the accurate detection of vtDR. This is particularly critical for identifying DMO, a vision-threatening complication of DR, as OCT is the gold standard for precise measurement of macular thickness and contour. Without treatment, patients with clinically significant DMO are twice more likely to experience visual loss [29, 30]. Kermany et al. showed success in training DLAs to detect DMO from OCT scans, with an AUC of 0.999 [31]. In our deep learning system, the inclusion of OCT improved our PPV from 43.8% to 67.9% while increasing overall accuracy and sensitivity. This dual-modality approach could reduce the likelihood of misdiagnoses and inappropriate referrals caused by false positives. Dual-modality systems have been demonstrated to be both feasible and accurate by Liu et al. who were able to detect referable DR with a sensitivity of 97.8% and AUC of 0.98, and DMO from OCT scans with a sensitivity of 91.3% and AUC of 0.94 [16]. These findings underscore the value of integrating OCT into AI-driven DR screening, demonstrating that dual-modality approaches could significantly improve both accuracy and clinical effectiveness.

TABLE 2 | Overall prevalence of DR by ophthalmologist diagnosis and mean visual acuity of the South Western Eye and Diabetes Deep Learning Algorithm (SWEDDLA) study participants.

Ophthalmologist diagnosis	***n* (%)**	
No DR	470 (63.2%)	
Mild NPDR	63 (8.4%)	
Moderate NPDR	119 (16.0%)	
Severe NPDR	19 (2.6%)	
PDR	17 (2.3%)	
Ungradable images	55 (7.4%)	
DMO	42 (7.8%)	
vtDR	63 (8.4%)	
Vision		
Mean VA logMAR; Snellen acuity	R: 0.1; 6/7.5	L: 0.1; 6/7.5

Abbreviations: DMO, diabetic macular oedema; DR, diabetic retinopathy; NPDR, non-proliferative diabetic retinopathy; PDR, proliferative diabetic retinopathy; VA, visual acuity; vtDR = vision-threatening diabetic retinopathy (severe NPDR/PDR and/or DMO).

Emerging evidence from other studies has shown that DLAs could be used in many ways to improve DR screening [32]. The most basic use would be as a decision-support tool, assisting non-ophthalmic clinicians in the grading of DR. Similarly, they could be integrated into existing DR programmes to assist in the triage of referrals and screening out those that do not need ophthalmologist review. This model of care has been successfully adopted in Portugal where the AI programme, Retmarker: grades fundus photographs to detect 'disease' or 'no disease' [33]. This is part of a two-step process in a Portuguese reading centre for DR screening, where all fundus photographs are first graded by Retmarker and any 'disease' photos are flagged for human grading. They found a 48.4% reduction in the human grading burden [33]. Other studies have shown that DLAs could be utilised as a complete replacement for human graders or as an autonomous referral triage tool, and have already been incorporated into clinical practice [34, 35]. A 2019 clinical validation study conducted in Zambia suggested that DLAs could provide an alternative solution for DR screening, especially in settings with little access to human expertise [36]. Sub-Saharan Africa has less than three ophthalmologists per million population, the lowest globally, contrasting with approximately 40 ophthalmologists per million population in Australia [36, 37]. Their AI programme obtained a sensitivity of 99.4% and AUC of 0.93 for the detection of vtDR, which is comparable to our deep learning system, and supports the potential application of DLAs in under-resourced settings [36]. An AI-driven DR screening programme is achievable, accurate and well-accepted as seen in a study published by Mingguang He's group, which tested end-user acceptability and integration of DLAs into clinic flow, finding that 96% of study participants were satisfied or very satisfied with an automated screening model [38].

The effectiveness of DLAs has been fundamentally limited by the scarcity of vtDR images, necessitating reliance on big datasets

TABLE 3 | Internal validation test characteristics of the fundus photograph and OCT algorithms.

Algorithm			
Characteristic	**Gradability of fundus photograph**	**Severe NPDR/PDR**	**DMO on OCT**
Accuracy	93.0%	94.0%	97.5%
Sensitivity	94.8%	90.3%	96.8%
Specificity	91.2%	97.7%	98.1%
PPV	91.3%	95.5%	98.1%
NPV	94.7%	95.5%	96.9%
AUC (95% CI)	0.98 (0.97–0.98)	0.99 (0.99–0.99)	0.99 (0.99–0.99)

Abbreviations: AUC, area under receiver operating characteristic curve; DMO, diabetic macular oedema; NPDR, non-proliferative diabetic retinopathy; NPV, negative predictive value; OCT, optical coherence tomography; PDR, proliferative diabetic retinopathy; PPV, positive predictive value.

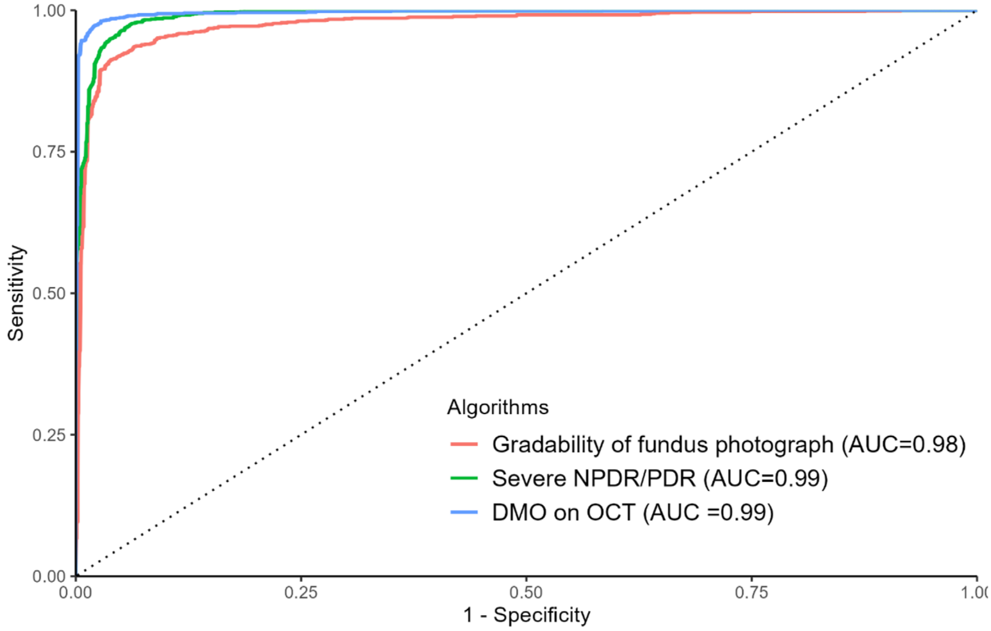

FIGURE 2 | Receiver operating characteristic curve of the deep learning algorithm for detection of gradable fundus photograph, severe non-proliferative and proliferative diabetic retinopathy, and diabetic macular oedema, in the internal validation dataset. AUC, area under receiver operating characteristic curve; DMO, diabetic macular oedema; NPDR, non-proliferative diabetic retinopathy; OCT, optical coherence tomography; PDR, proliferative diabetic retinopathy.

TABLE 4 | External validation test characteristics of the fundus photograph and OCT algorithms.

Algorithm				
Characteristic	**Gradability of fundus photograph**	**Severe NPDR/PDR**	**DMO on OCT**	**vtDR**
Accuracy	94.5%	94.2%	97.7%	95.2%
Sensitivity	95.3%	88.1%	81.4%	92.7%
Specificity	88.7%	94.5%	98.6%	95.5%
PPV	98.4%	43.8%	77.4%	67.9%
NPV	72.0%	99.4%	98.9%	99.2%
AUC (95% CI)	0.92 (0.90–0.94)	0.91 (0.87–0.96)	0.90 (0.85–0.95)	0.94 (0.91–0.97)

Abbreviations: AUC, area under receiver operating characteristic curve; DMO, diabetic macular oedema; NPDR, non-proliferative diabetic retinopathy; NPV, negative predictive value; OCT, optical coherence tomography; PDR, proliferative diabetic retinopathy; PPV, positive predictive value; vtDR, vision-threatening diabetic retinopathy (severe NPDR/PDR and/or DMO).

that are difficult to obtain. In the SWEDDLA study, data augmentation techniques were employed to generate high-quality synthetic images of severe NPDR and PDR, which enhanced the training dataset and improved algorithm performance. Without data augmentation, our deep learning system demonstrated reduced sensitivity to vision-threatening disease, with overfitting due to fewer positive cases and diminished performance in the external validation dataset. A significant challenge lies in the need for meticulous labelling of training images to ensure a high-performing DLA. For our system, we undertook the labour-intensive task of regrading and labelling all training images, which were obtained from publicly available datasets. This process required repeated refinements to achieve optimal internal validation test characteristics, delaying the deployment of satisfactory DLAs for real-world testing. This limitation could be mitigated by adopting DLAs equipped with self-supervised learning (SSL), a more efficient, time-saving and cost-effective approach [39, 40]. Zhou et al. have developed a new SSL-based foundation model for retinal images (RETFound) and systematically evaluated its performance in DR screening. Using unlabelled images comprising 904170 fundus photographs and 736442 OCT scans, they were able to achieve an AUC of 0.82 and 0.74 on the publicly available IDRID and MESSIDOR-2 datasets [40]. Another recent advance has been in generative adversarial networks (GANs), which have enabled the automatic generation of photorealistic images that can be manipulated with arbitrary grading and lesion information [41, 42]. Thus, large-scale generated data could be used for more meaningful augmentation to train DR grading and lesion segmentation systems [42]. The use of GANs in DR screening has shown promise, with a 2024 paper by Sebastian et al. demonstrating an accuracy of 98.3%, sensitivity of 97.3% and specificity of 99.2% for

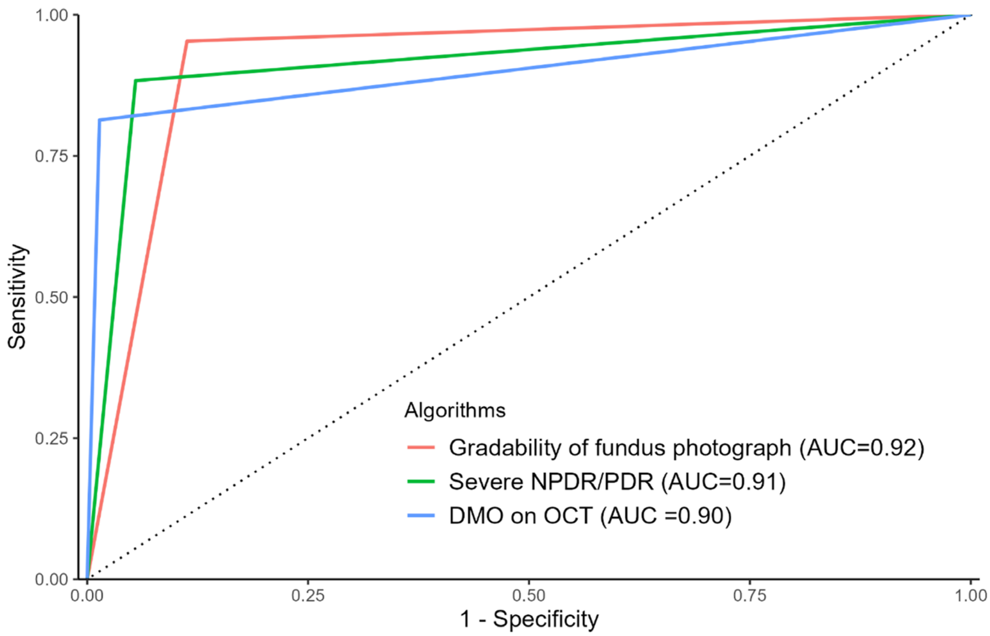

FIGURE 3 | Receiver operating characteristic curve of the deep learning algorithm for detection of gradable fundus photograph, severe non-proliferative and proliferative diabetic retinopathy and diabetic macular oedema, in the external validation dataset. AUC, area under receiver operating characteristic curve; DMO, diabetic macular oedema; NPDR, non-proliferative diabetic retinopathy; OCT, optical coherence tomography; PDR, proliferative diabetic retinopathy.

the extraction of retinal vasculature lesions from fundus photographs with DR [43]. Although, there has been much progress in the use of DLAs in ophthalmology over the last ten years, the use of GANs is less pronounced [44]. It represents an exciting future direction for the field, with further studies being required to determine whether GAN techniques will improve the diagnostic performance of DLAs in real-world testing [41, 44].

Our study has several strengths. First, our deep learning system was evaluated in a novel population that included paediatric participants, suffering from a high burden of DR, in comparison to the national average. Second, this was a prospective study enabling observations to be made into the feasibility of using DLAs at the point-of-care in non-ophthalmic settings. Third, our system utilised a dual-modality approach that included OCT to diagnose DMO, enabling far greater accuracy in diagnosing vtDR. Fourth, we applied a rigorous and consistent screening approach to ensure the quality of each training image used to create our deep learning system. Fifth, we reported a range of composite outcomes clinically relevant to real-world screening programmes, including vtDR, DMO and fundus gradability. Sixth, we reported a high fundus and OCT gradability percentage. One limitation of this study was that it was not population-based, having been conducted across multiple primary and tertiary care sites. In addition, the number of paediatric participants was lower than adults, making it challenging to draw robust conclusions about the paediatric cohort. Finally, our overall sample size was smaller than that of more recent prospective studies. A difference of the SWEDDLA study was that we used vtDR as our primary clinical endpoint, unlike most that focused on referable DR. This decision was based on the study's location in a district of Australia with one of the highest prevalence rates of diabetes and corresponding levels of vtDR. We believed this measure was more practical for optimising onward referral pathways and prioritising patients who required urgent treatment, in the already overburdened, public ophthalmology clinics of SWS. Additionally, we generated a binary graph to represent the AUC in the external validation dataset. This was based on the programme's default operating point threshold (≥ 0.5) for determining the presence or absence of disease, resulting in a comparison between two discrete variables: the DLAs' diagnosis and the ophthalmologists' diagnosis. Future research will focus on determining the optimal operating point in the external validation dataset for each DLA used in this study. The SWEDDLA study utilised both a table-top and a portable retinal camera for image capture, suggesting that future research could investigate additional retinal cameras with varying image characteristics and participant cohorts. Future work is being conducted to investigate participant satisfaction with retinal imaging as part of SWS multidisciplinary diabetes and GP clinics. We are also exploring the technical feasibility, cost-effectiveness and impact on clinic flow of integrating our deep learning system into the standard DR screening pathway.

In summary, the SWEDDLA study shows a clinically acceptable deep learning system that compares well to existing algorithms in terms of AUC, accuracy, sensitivity and specificity. It was developed to address the specific needs of our local population, using easily accessible computers, software and techniques and publicly available datasets. It has real-world applicability with robust test characteristics, validated in both adult and paediatric participants. This highlights the ability of deep learning systems to deliver specialist-level diagnostics in diabetes or GP clinics, which could increase DR screening rates overall. The addition of OCT enhanced the detection of vtDR and could optimise onward referral pathways. The system described in this study is poised to support a novel model of care for DR screening in SWS.

Acknowledgements

We give thanks to Dr. Sheng Chiong Hong (Director and Co-Founder, oDocs Eye Care) and Mr. Glenn Linde (CTO, oDocs Eye Care) for assistance with the creation of the deep learning system used in this study [Correction added on 30 July 2025, after first online publication: In the Acknowledgements section, 'Graham Linde' has been corrected to 'Glenn Linde'; The institutional affiliations of Dr Hong and Mr Linde have been added in this version.]. Open access publishing facilitated by Western Sydney University, as part of the Wiley - Western Sydney University agreement via the Council of Australian University Librarians.

Conflicts of Interest

S.K. has a patent pending for the deep learning system described in this paper. The other authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

1. L. F. Nakayama, L. Z. Ribeiro, M. B. Gonçalves, et al., "Diabetic Retinopathy Classification for Supervised Machine Learning Algorithms," *International Journal of Retina and Vitreous* 8, no. 1 (2022): 1.

2. D. Australia, *The State of the Nation: The Diabetes Epidemic in Australia* (Diabetes Australia, 2024).

3. S. Tatulashvili, G. Fagherazzi, C. Dow, R. Cohen, S. Fosse, and H. Bihan, "Socioeconomic Inequalities and Type 2 Diabetes Complications: A Systematic Review," *Diabetes & Metabolism* 46, no. 2 (2020): 89–99.

4. G. Liew, V. W. Wong, M. Saw, et al., "Profile of a Population-Based Diabetic Macular Oedema Study: The Liverpool Eye and Diabetes Study (Sydney)," *BMJ Open* 9, no. 1 (2019): e021884.

5. G. Liew, T. Tsang, B. Marshall, et al., "Proportion of People With Diabetic Retinopathy and Macular Oedema Varies by Ethnicity in a Tertiary Retinal Clinic in Australia: Findings From the Liverpool Eye and Diabetes Study (LEADS)," *BMJ Open* 13, no. 2 (2023): e055404.

6. C. J. Flaxel, R. A. Adelman, S. T. Bailey, et al., "Diabetic Retinopathy Preferred Practice Pattern," *Ophthalmology* 127, no. 1 (2020): P66–P145.

7. Ophthalmologists RANZCO, *RANZCO Screening and Referral Pathway for Diabetic Retinopathy* (Royal Australian and New Zealand College of Ophthalmologists, 2019).

8. L. Crossland, D. Askew, R. Ware, et al., "Diabetic Retinopathy Screening and Monitoring of Early Stage Disease in Australian General Practice: Tackling Preventable Blindness Within a Chronic Care Model," *Journal of Diabetes Research* 2016 (2016): 1–7.

9. S. Mazumdar, N. Bagheri, S. Chong, I. S. McRae, B. Jalaludin, and F. Girosi, "Diabetes and the Use of Primary Care Provider Services in Rural, Remote and Metropolitan Australia," *Rural and Remote Health* 21, no. 3 (2021): 5844, https://doi.org/10.22605/RRH5844.

10. M. J. G. Watson, P. J. McCluskey, J. R. Grigg, Y. Kanagasingam, J. Daire, and M. Estai, "Barriers and Facilitators to Diabetic Retinopathy Screening Within Australian Primary Care," *BMC Family Practice* 22, no. 1 (2021): 239, https://doi.org/10.1186/s12875-021-01586-7.

11. J. Foreman, S. Keel, J. Xie, P. Van Wijngaarden, H. R. Taylor, and M. Dirani, "Adherence to Diabetic Eye Examination Guidelines in Australia: The National Eye Health Survey," *Medical Journal of Australia* 206, no. 9 (2017): 402–406.

12. D. J. McCarty, C. L. Fu, C. A. Harper, H. R. Taylor, and C. A. McCarty, "Five-Year Incidence of Diabetic Retinopathy in the Melbourne Visual Impairment Project," *Clinical & Experimental Ophthalmology* 31, no. 5 (2003): 397–402.

13. R. J. Tapp, P. Z. Zimmet, C. A. Harper, et al., "Diabetes Care in an Australian Population," *Diabetes Care* 27, no. 3 (2004): 688–693, https://doi.org/10.2337/diacare.27.3.688.

14. J. Xie, A.-L. Arnold, J. Keeffe, et al., "Prevalence of Self-Reported Diabetes and Diabetic Retinopathy in Indigenous Australians: The National Indigenous Eye Health Survey," *Clinical & Experimental Ophthalmology* 39, no. 6 (2011): 487–493.

15. S. Natarajan, A. Jain, R. Krishnan, A. Rogye, and S. Sivaprasad, "Diagnostic Accuracy of Community-Based Diabetic Retinopathy Screening With an Offline Artificial Intelligence System on a Smartphone," *JAMA Ophthalmology* 137, no. 10 (2019): 1182–1188.

16. R. Liu, Q. Li, F. Xu, et al., "Application of Artificial Intelligence-Based Dual-Modality Analysis Combining Fundus Photography and Optical Coherence Tomography in Diabetic Retinopathy Screening in a Community Hospital," *Biomedical Engineering Online* 21, no. 1 (2022): 47.

17. G. Linde, W. De Rodrigues Souza Jr, R. Chalakkal, H. V. Danesh-Meyer, B. O'Keeffe, and S. Chiong Hong, "A Comparative Evaluation of Deep Learning Approaches for Ophthalmology," *Scientific Reports* 14, no. 1 (2024): 21829.

18. https://www.medicmind.tech/.

19. National Health Service Scotland, *Clinical Guideline: Diabetic Retinopathy Grading Scheme Scotland* (National Health Service, 2021).

20. R. K. Malhotra and A. Indrayan, "A Simple Nomogram for Sample Size for Estimating Sensitivity and Specificity of Medical Tests," *Indian Journal of Ophthalmology* 58, no. 6 (2010): 519–522.

21. P. H. Scanlon, "The English National Screening Programme for Diabetic Retinopathy 2003–2016," *Acta Diabetologica* 54, no. 6 (2017): 515–525.

22. V. Gulshan, L. Peng, M. Coram, et al., "Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs," *Journal of the American Medical Association* 316, no. 22 (2016): 2402–2410.

23. P. Ruamviboonsuk, R. Tiwari, R. Sayres, et al., "Real-Time Diabetic Retinopathy Screening by Deep Learning in a Multisite National Screening Programme: A Prospective Interventional Cohort Study," *Lancet Digital Health* 4, no. 4 (2022): e235–e244, https://doi.org/10.1016/S2589-7500(22)00017-6.

24. E. Ipp, D. Liljenquist, B. Bode, et al., "Pivotal Evaluation of an Artificial Intelligence System for Autonomous Detection of Referrable and Vision-Threatening Diabetic Retinopathy," *JAMA Network Open* 4, no. 11 (2021): e2134254.

25. A. Y. Lee, R. T. Yanagihara, C. S. Lee, et al., "Multicenter, Head-to-Head, Real-World Validation Study of Seven Automated Artificial Intelligence Diabetic Retinopathy Screening Systems," *Diabetes Care* 44, no. 5 (2021): 1168–1175.

26. S. Joseph, J. Selvaraj, I. Mani, et al., "Diagnostic Accuracy of Artificial Intelligence-Based Automated Diabetic Retinopathy Screening in Real-World Settings: A Systematic Review and Meta-Analysis," *American Journal of Ophthalmology* 263 (2024): 214–230.

27. A. Shah, W. Clarida, R. Amelon, et al., "Validation of Automated Screening for Referable Diabetic Retinopathy With an Autonomous Diagnostic Artificial Intelligence System in a Spanish Population," *Journal of Diabetes Science and Technology* 15, no. 3 (2021): 655–663.

28. J. Cuadros, "The Real-World Impact of Artificial Intelligence on Diabetic Retinopathy Screening in Primary Care," *Journal of Diabetes Science and Technology* 15, no. 3 (2021): 664–665.

29. Y. S. Yuen, J. S. Gilhotra, M. Dalton, et al., "Diabetic Macular Oedema Guidelines: An Australian Perspective," *Journal of Ophthalmology* 2023, no. 1 (2023): 1–22.

30. Photocoagulation for Diabetic Macular Edema, "Early Treatment Diabetic Retinopathy Study Report Number 1 Early Treatment Diabetic Retinopathy Study Research Group," *Archives of Ophthalmology* 103, no. 12 (1985): 1796–1806.

31. D. S. Kermany, M. Goldbaum, W. Cai, et al., "Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning," *Cell* 172, no. 5 (2018): 1122–1131.e9.

32. E. R. Pedersen, J. A. Cuadros, M. L. Khan, et al., "Redesigning Clinical Pathways for Immediate Diabetic Retinopathy Screening Results," *NEJM Catalyst Innovations in Care Delivery* 2 (2021).

33. L. Ribeiro, C. M. Oliveira, C. Neves, J. D. Ramos, H. Ferreira, and J. Cunha-Vaz, "Screening for Diabetic Retinopathy in the Central Region of Portugal. Added Value of Automated 'Disease/No Disease' Grading," *Ophthalmologica* 233, no. 2 (2014): 96–103.

34. A. E. Rajesh, O. Q. Davidson, C. S. Lee, and A. Y. Lee, "Artificial Intelligence and Diabetic Retinopathy: AI Framework, Prospective Studies, Head-to-Head Validation, and Cost-Effectiveness," *Diabetes Care* 46, no. 10 (2023): 1728–1739.

35. A. Grzybowski, P. Singhanetr, O. Nanegrungsunk, and P. Ruamviboonsuk, "Artificial Intelligence for Diabetic Retinopathy Screening Using Color Retinal Photographs: From Development to Deployment," *Ophthalmology and Therapy* 12, no. 3 (2023): 1419–1437.

36. V. Bellemo, Z. W. Lim, G. Lim, et al., "Artificial Intelligence Using Deep Learning to Screen for Referable and Vision-Threatening Diabetic Retinopathy in Africa: A Clinical Validation Study," *Lancet Digital Health* 1, no. 1 (2019): e35–e44.

37. P. Allen, B. Jessup, S. Khanal, V. Baker-Smith, K. Obamiro, and T. Barnett, "Distribution and Location Stability of the Australian Ophthalmology Workforce: 2014–2019," *International Journal of Environmental Research and Public Health* 18, no. 23 (2021): 12574.

38. S. Keel, P. Y. Lee, J. Scheetz, et al., "Feasibility and Patient Acceptability of a Novel Artificial Intelligence-Based Screening Model for Diabetic Retinopathy at Endocrinology Outpatient Services: A Pilot Study," *Scientific Reports* 8, no. 1 (2018): 4330.

39. J. Zhang, S. Lin, T. Cheng, et al., "RETFound-Enhanced Community-Based Fundus Disease Screening: Real-World Evidence and Decision Curve Analysis," *NPJ Digital Medicine* 7, no. 1 (2024): 108.

40. Y. Zhou, M. A. Chia, S. K. Wagner, et al., "A Foundation Model for Generalizable Disease Detection From Retinal Images," *Nature* 622, no. 7981 (2023): 156–163.

41. A. You, J. K. Kim, I. H. Ryu, and T. K. Yoo, "Application of Generative Adversarial Networks (GAN) for Ophthalmology Image Domains: A Survey," *Eye and Vision* 9, no. 1 (2022): 6.

42. G. Lim, P. Thombre, M. Lee, and W. Hsu, "Generative Data Augmentation for Diabetic Retinopathy Classification," in 2020 *IEEE 32nd International Conference on Tools With Artificial Intelligence (ICTAI)*, (IEEE, 2020), 1096–1103.

43. A. Sebastian, O. Elharrouss, S. Al-Maadeed, and N. Almaadeed, "GAN-Based Approach for Diabetic Retinopathy Retinal Vasculature Segmentation," *Bioengineering* 11, no. 1 (2024): 4.

44. Y. Zhou, B. Wang, X. He, S. Cui, and L. Shao, "DR-GAN: Conditional Generative Adversarial Network for Fine-Grained Lesion Synthesis on Diabetic Retinopathy Images," *IEEE Journal of Biomedical and Health Informatics* 26, no. 1 (2022): 56–66.

[Correction added on 30 July 2025, after first online publication: A new reference 18 has been added, and subsequent references have been renumbered accordingly]